A Geometry-Based Computational Assessment of Parcel-Level Redevelopment Feasibility in Seoul

Sanghoon Byun[1] and Dongjun Park[2], Bumjun Kang[3]
[1,2,3]Seoul National Uvinersity, Laus.(Lab. Architectural&Urban Space)
[1]bsh960flash@snu.ac.kr , 0009-0009-0741-1973
[2]gbjun7333@snu.ac.kr, 0009-0006-0959-0002
[3]bumjoon@snu.ac.kr, 0000-0002-8551-7652

Abstract. Seoul's urban fabric is characterized by highly fragmented and historically inefficient parcel layouts that constrain redevelopment. In response, the city has promoted Small Residential Redevelopment Models (SRRMs) as an institutional strategy. However, systematic tools for assessing the geometric feasibility of these parcels remain underdeveloped, particularly those capable of integrating complex local building codes and economic land value standards. This study introduces an automated, geometry-based computational framework to evaluate parcel efficiency through a novel metric termed the Parcel Layout Efficiency (PLE) Index. The model quantifies four geometric indices—Shape, Topography, Buildable Area, and Road Access—each weighted according to user-defined parameters. Applied to more than 42,000 urban blocks across Seoul, the analysis reveals substantial quantitative differences between historically unplanned and modern planned areas, thereby validating the model's analytical robustness. This research provides a foundational framework for pre-project feasibility analysis, offering a scalable tool for data-driven evaluation of urban redevelopment potential and contributing to the advancement of computational methods in urban morphology and design decision-making.

Keywords. Geometry-Based Automation, Parcel Layout Evaluation, Feasibility Test, Urban Regeneration, Computational Design, Small Residential Redevelopment, Seoul, Rhino-Grasshopper

1. Introduction

1.1. BACKGROUND AND PROBLEM STATEMENT

Seoul's urban fabric is fragmented by historic organic developments, yielding irregular or inefficient parcel layouts, such as landlocked or flag-shaped parcels. These parcels are a major factor hindering development. In most cases, a development on such a single, irregular parcel may not be acceptable (Korean Building Act, Article 44). It is significantly inefficient to develop such parcels in terms of the systematic land use management (Lee et al., 2009)., ultimately detracting from the parcel's value (Villanueva & Colombo, 2021).

To address this problem, "Small-scale Residential Redevelopment Models (SRRMs)" are being institutionally encouraged in Seoul (Seoul Metropolitan Government Ordinance on Urban and Residential Environment Maintenance; KRIHS, 2023). However, existing tools are inadequate for evaluating the parcel layout feasibility of these redevelopment project sites. There is a lack of objective indices that integrally reflect complex local building regulations and economic land value standards (Official Standard Land Price).

1.2. RESEARCH OBJECTIVES AND SCOPE

The objective of this paper is to develop an automated, geometry-based model, the Parcel Layout Efficiency (PLE) index, to evaluate the relative efficiency levels of these parcel layouts to apply SRRM projects. The research question is: "Can we computationally model parcel efficiency by integrating geometric constraints with Korea's specific legal (building regulations) and economic (official land appraisal) standards?"

To this end, we propose a computational framework developed using the Rhino Python API within the Grasshopper environments, linked with GIS. This framework focuses on evaluating the geometric aspects of parcel layouts, automating the assessment of four core indices—Shape, Topography, Buildable Area, and Road Access—which reflect the standards of the Official Standard Land Price and building regulations.

This study is a preparatory step toward the ultimate goal of reviewing the feasibility of specific "Small-scale Residential Redevelopment Model" projects before implementation. Therefore, prior to detailed verification of specific project models, we conduct a foundational analysis by applying the developed PLE model to over 42,000 blocks in Seoul to identify macro-trends. Specifically, we validate the model's core effectiveness by comparing the PLE indices of 'historically unplanned residential areas' and modern 'planned residential areas', utilizing the classifications from the Seoul Research Institute's (2009) <Study on the Urban Form of Seoul> report.

2. Related Work

2.1. PARCEL SHAPE AND LAND VALUE

The physical characteristics of parcels, especially their shape, are a key factor in determining land's economic value. Villanueva & Colombo (2021) argued that the geometric characteristics of parcels directly affect land use efficiency and development potential, which is ultimately reflected in land value.

Korea's Official Standard Land Price (OSLP) appraisal system is a representative case that institutionalizes this relationship. According to the OSLP survey and evaluation standards (MOLIT, 2021), parcel shape is reflected in land price calculation as one of the 'lot conditions' and is classified into eight types. Studies (e.g., Park, 2023) confirm that in Korean appraisal practice, regular shapes are systematically valued higher than irregular ones. Ham & Kim (2024) also noted this practice, pointing out issues of subjectivity and proposing deep-learning models for classification.

This valuation perspective aligns with urban morphology research (Kropf, 1996), where landlocked or irregular parcels lead to construction constraints (Colwell & Scheu, 1998), causing land use inefficiency and value depreciation (Lin, Su & Li, 2023). This aligns perfectly with the concept of 'inefficient parcel layout' defined in the `Introduction`, highlighting the need for quantitative evaluation.

2.2. LIMITATIONS OF PRIOR WORK AND THIS STUDY'S CONTRIBUTION

Existing research on quantifying parcel layout efficiency shows three clear limitations in evaluating the parcel layout feasibility for the "Small-scale Residential Redevelopment Models" mentioned in the `Introduction`.

First, the content of the indices fails to reflect the real-world development context. Prior studies have relied on simple 2D Shape Indices (Demetriou et al., 2013; Arslan, 2021), failing to incorporate the physical and institutional context of building regulations or road access. This leads to structural limitations in assessing a parcel's actual value, such as rating a geometrically 'efficient' but practically unbuildable parcel as 'superior' (Arslan, 2021).

Second, the analysis scale is disconnected from policy reality. Existing research (Zhang et al., 2023) has been confined to fixed units of 'individual parcels' (Hong & Lee, 2022) or the 'entire block' (Li & Quan, 2024). This leaves a gap in analytical frameworks capable of measuring the efficiency of 'Groups of Parcels', which are flexibly formed based on real-world needs in projects like "Small-scale Residential Redevelopment Models".

Third, the evaluation method is stuck at an 'absolute' standard. Existing indices (Demetriou et al., 2013) attempt to measure absolute 'excellence' applied uniformly to all parcels, such as proximity to a predefined 'ideal shape'. However, parcel efficiency should be approached from the perspective of 'relative efficiency'—evaluating how efficiently a parcel is laid out compared to the maximum potential of its containing block, not against an 'ideal shape'. Such a relative comparison model is absent in prior research.

These limitations clearly show the need for an index that (1) reflects the real-world institutional context (value, regulations), (2) aligns with the policy scale (parcel groups), and (3) can evaluate 'relative efficiency' against the block's potential. This study proposes the PLE index to overcome these three gaps.

3. Method

This study develops the Parcel Layout Efficiency (PLE) evaluation system, which automates geometric evaluation, to quantitatively measure the 'inefficient parcel layout' (i.e., layouts that detract from parcel value) defined in the `Introduction`.

This system is based on the computational framework proposed in the `Introduction`. That is, it links a framework developed using the Rhino Python API in a Grasshopper environment with GIS, focusing on the geometric aspects of parcel layouts. It provides a framework that automatically evaluates efficiency across various scales, from individual parcels to entire blocks, based on four core indices (Shape, Topography, Buildable Area, Road Access) that combine Seoul's GIS data with Korean building regulations and Official Standard Land Price standards.

3.1. EVALUATION FRAMEWORK OVERVIEW

The entire workflow of the PLE evaluation system proposed in this study consists of four stages, as shown in Figure 1

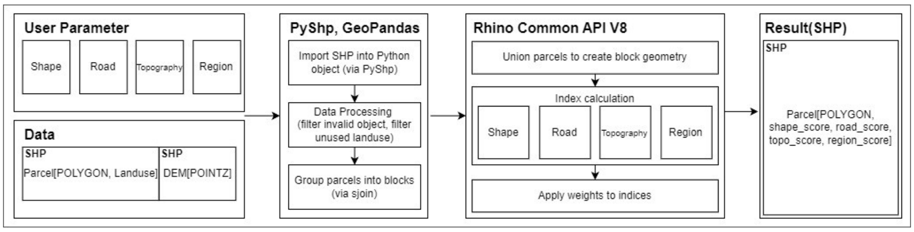

Figure 1. Flowchart of the PLE evaluation system

Rationales for the Four Indices in the PLE Framework

The computational logic of each index is grounded in four reference frameworks: (1) Korea's Official Standard Land Price appraisal criteria (MOLIT, 2025), (2) the Korean Building Act and its enforcement decree, (3) geometric shape parameters from prior research (Demetriou et al., 2013), and (4) tacit design knowledge from architectural practice. These text-based legal and economic standards were translated into geometric logic to reflect real-world development context.

1. Shape Index: The three sub-indices of the Shape Index reflect concepts from both the appraisal criteria and prior research. It implements the "available area ratio" used in the appraisal standard to identify "irregular shapes." It also adapts the Compactness parameter from Demetriou et al. (2013). Finally, it combines the appraisal standard's "regular shape" definition (aspect ratio near 1:1.1) with Demetriou et al.'s Boundary Points parameter (four vertices as optimal).

2. Topography Index: This index quantifies the slope by following the topography classification in the MOLIT (2025) guidelines. In the Korean appraisal system, land is categorized into "Flat," "Gentle Slope," or "Steep Slope," as these factors directly determine construction costs and earthwork feasibility. The model calculates the average gradient of the parcel to reflect these physical and economic limitations.

3. Road Access Index: This index integrates the "Road Contact" criteria from the appraisal guidelines (MOLIT, 2025) and Article 44 (Relationship between Building Site and Road) of the Building Act. It evaluates both the economic value derived from road width (e.g., Broad, Medium, or Narrow roads) and the legal requirement for development permission, which necessitates a minimum 2-meter contact with a road.

4. Buildable Area Index: Rather than simple gross area, this index calculates the net developable volume by parameterizing Article 46 (Designation of the Building Line) and Article 58 (Clearance within the Site) of the Building Act. The algorithm accounts for mandatory setbacks from the "Building Line" and "Adjacent Boundary Lines," which significantly reduce the usable area in fragmented or narrow parcels, thereby reflecting the site's practical economic capacity.

3.2. FOUR CORE INDICES FOR PLE

The PLE index independently evaluates the efficiency of parcel layouts from four perspectives(1) Shape, (2) Topography, (3) Buildable Area, and (4) Road Access and calculates a weighted sum. Each index is normalized to a value between 0 and 1.

3.2.1. Shape Index, S_{shape}

Shape Index (S_{shape}) evaluates 'relative efficiency' as the ratio of the average shape score of individual parcels to the shape score of the entire block (undivided stat.

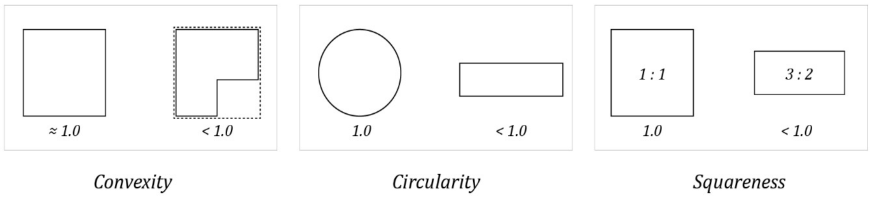

Figure 2. Example of Shape Index sub-elements.

$$S_{shape} = \frac{\frac{1}{n}\sum_{i=1}^{n} S_{shape}(i)}{S_{shape}(block)}$$

Where:

- n: Total number of parcels within the block
- $S_{shap}\ (i)$: Shape score of individual parcel i (weighted sum of sub-indices below)
- $S_{shape}(block)$: Shape score of the entire block (undivided state) (weighted sum of sub-indices below))

Each shape score ($S_{sha}\ (i)$ or $S_{shape}(block)$) is calculated as a weighted sum of three sub-indices:

$$S_{shape}(entity) = \\ w_{conv} \cdot S_{conv}(entity) + w_{circ} \cdot S_{circ}(entity) + w_{sq} \cdot S_{sq}(entity)$$

(where, $w_{conv} + w_{circ} + w_{sq} = 1$)

- S_{conv}: Convexity score(Area/Convex Hull Area)
- S_{circ}: Circularity score ($4\pi \cdot \text{Area}/\text{Perimeter}^2$)
- S_{sq}: Squareness score (aspect ratio and vertex penalty)

3.2.2. Topography Index, S_{topo}

The Topography Index ((S_{topo})) evaluates how much lower the average relief of subdivided parcels is compared to the relief of the entire block.

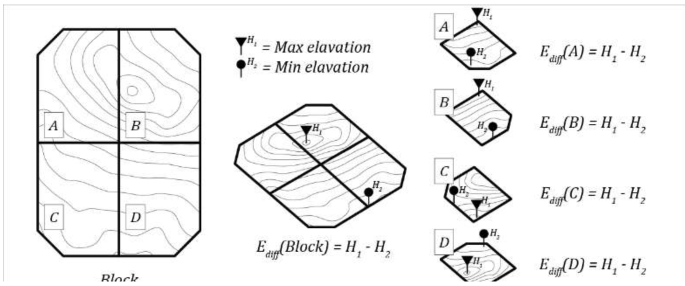

Figure 3. Conceptual diagram of the Topography Index (S_{topo}) evaluation.

$$S_{topo} = 1 - \frac{\frac{1}{n}\sum_{i=1}^{n} E_{diff}(i)}{E_{diff}(block)}$$

Where:

- n: Total number of parcels within the block
- $E_{diff}(i)$: Relief of individual parcel i(max elevation - min elevation within parcel)
- $\frac{1}{n}\sum_{i=1}^{n} E_{diff}(i)$: Average relief of individual parcels within the block
- $E_{diff}(block)$: Relief of the entire block (max elevation - min elevation within the block)

3.2.3. Buildable Region Index, S_{region}

The Buildable Region Index (S_region) evaluates 'relative efficiency' as the ratio of the 'sum' of buildable areas of currently subdivided parcels to the total buildable area of the entire block (undivided state).

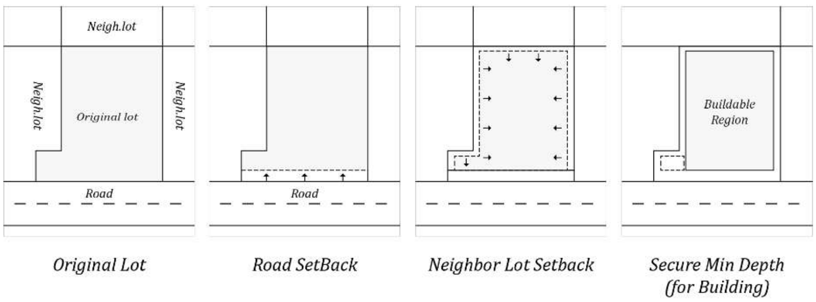

Figure 4. Example of Buildable Area calculation process.

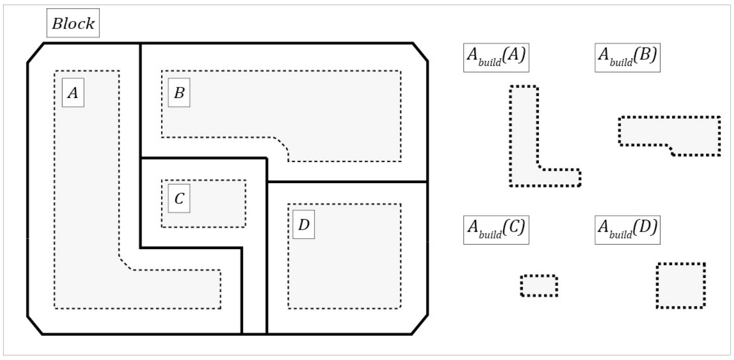

Figure 5. Conceptual diagram of the Buildable Region Index.

$$S_{region} = \frac{\sum_{i=1}^{n} A_{build}(i)}{A_{build}(block)}$$

Where:

- n: Total number of parcels within the block
- $A_{build}(i)$: Buildable area of individual parcel i (area after applying regulatory setbacks and minimum width requirements)
- $\sum_{i=1}^{n} A_{build}(i)$: Sum of buildable areas of all individual parcels within the block
- $A_{build}(block)$: Total buildable area of the entire block (undivided state)

3.2.4. Road Access Index, S_{road}

The Road Access Index (S_{road}) evaluates 'relative efficiency' as the ratio of the average road access score of currently subdivided parcels to the maximum possible road access score the block can achieve.

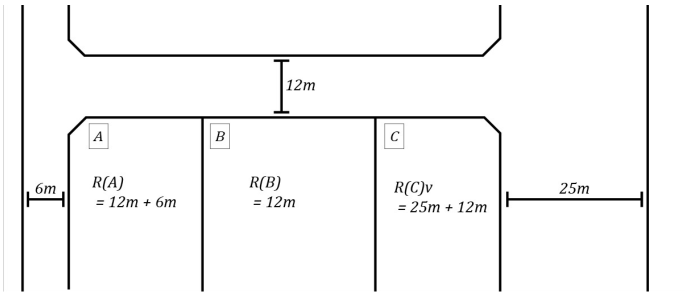

Figure 6. Conceptual diagram of the Road Access Index (S_{road}) evaluation.

$$S_{road} = \frac{\frac{1}{n}\sum_{i=1}^{n} R(i)}{R_{block}}$$

Where:

- n: Total number of parcels within the block
- $R(i)$: Road access score of individual parcels i (multiplier value assigned according to 12 types in the land price ratio table)
- $\frac{1}{n}\sum_{i=1}^{n} R(i)$: Average road access score of individual parcels within the block
- R_{block}: Maximum road access score the entire block (undivided state) can achieve

3.3. OVERALL PLE SCORE CALCULATION AND USER CUSTOMIZATION

The final Parcel Layout Efficiency (PLE) index is calculated through a 'weighted sum' of the four core indices (S).

$$PLE = (w_{sha} \cdot S_{shape}) + (w_{topo} \cdot S_{topo}) + (w_{region} \cdot S_{region}) + (w_{road} \cdot S_{road})$$

- $w_{shape}, w_{topo}, w_{region}, w_{road}$: User-defined weights for each index (constraint, $w_{shape} + w_{topo} + w_{region} + w_{road} = 1$)
- $S_{shape}, S_{topo}, S_{region}, S_{road}$: Four block-level relative efficiency indices calculated above (values between 0-1)

Importantly, as evident from the formula, the PLE index evaluates 'Relative Efficiency'. It is not an absolute value index measuring the worth of subdivided parcel outcomes (V_i), but rather a ratio index assessing how efficiently the current 'subdivision action' has been performed relative to the block's total potential (V_{block}) before subdivision.

The PLE index ranges between 0 and 1, where values closer to 1 indicate that the current parcel layout maximally utilizes the block's potential value, while values closer to 0 indicate severe layout inefficiency and high need for integrated development (re-subdivision).

3.4. APPLICATION TO ANALYSIS SITES

The developed PLE evaluation system was applied to 902,001 target parcels across Seoul. These parcels were grouped into a total of 42,000 blocks. We selected two residential zones (Type 1 and Type 2 General Residential), which are the most common in Seoul. Within these zones, eight 'historically unplanned residential areas' and eight 'planned residential areas' were selected to compare the PLE index values by land use type and development history.

4. Result

Of the 42,000 blocks initially identified, certain cases were excluded from the analysis. Single-parcel blocks (where PLE scores inherently converge to 1) and large-scale apartment complexes (inadequately represented in the data) were removed. This filtering process resulted in a final dataset of 27,491 blocks.

For this foundational analysis, all four indices were assigned equal weights

($w_{shape} = w_{topo} = w_{region} = w_{road} = 0.25$).

Furthermore, as this study is intended to be applicable to Small-scale Residential Redevelopment Models (SRM), the user-configurable parameters, including building regulations (e.g., setbacks), were set based on standards for residential use.

The analysis of these blocks confirmed a significant quantitative difference between historically unplanned and modern planned areas, validating the model's effectiveness. High-ranked blocks (high PLE) were concentrated in planned development districts (e.g., Gangnam-gu, Seocho-gu), while low-ranked blocks were predominantly located in historic urban cores and hillside residential areas (e.g., Gwanak-gu, Jongno-gu).

This divergence is not just in the final PLE score but is evident in the individual indices. For example, the Shape Index (S_{shape}) clearly visualizes this difference. Figure 7 (a) shows a planned area in Gangnam-gu, characterized by regular, rectilinear parcels that result in high shape scores (dark red). In contrast, Figure 7 (b) shows a historically unplanned area in Jongno-gu, where irregular and fragmented parcels result in very low shape scores (light/white).

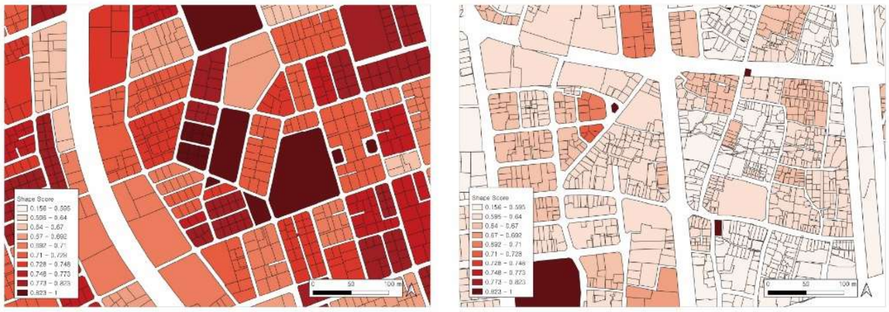

Figure 7. Comparison of Shape Index (S_{shape}) scores. (a) High scores in a planned area (Gangnam-gu). (b) Low scores in an unplanned area (Jongno-gu).

While some blocks had uniformly low scores, many low-PLE blocks were characterized by critical deficiencies in one or two specific indices (e.g., S_{topo} or S_{road}), suggesting targeted areas for improvement rather than uniform poor performance across all metrics.

5. Conclusion

This study proposed an automated PLE evaluation framework, using Rhino/Grasshopper and GIS, to assess parcel layout efficiency in Seoul. It integrates Korean legal (regulations) and economic (Official Standard Land Price) standards by evaluating four indices (Shape, Topography, Buildable Area, Road Access).

Applied to 27,491 blocks, the analysis revealed a wide PLE score range (0.917 to 0.052). This confirmed a significant efficiency gap between planned (high-ranked) and historic/hillside (low-ranked) areas, with low scores often driven by specific index failures (e.g., $\mathrm{S_{topo}}$ or $\mathrm{S_{road}}$).

This study provides a foundational analysis for identifying macro-trends in parcel layout efficiency across Seoul. The framework may serve as a tool for evaluating alternative configurations in future re-parceling simulations. Additionally, further validation is needed to examine statistical correlations between PLE scores and actual land prices or development density such as floor area ratio (FAR).

6. Acknowledgements

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Gemini (Google, 2024) was used for manuscript refinement and translation.

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